Akoizo Original Practice - Form 1 - Answer key

Division B | Ping Pong Parachute | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Three timers read 12.41, 12.56, and 12.30 s. What is the official time?

A. 12.41 s

B. 12.56 s

C. 12.42 s

D. 12.30 s

Correct: A

2. (2 points) Which deployment method is allowed?

Unit: Deployment and separation.
Worked-case data: Video shows a payload leaving the rocket 0.3 s after launch, well before apogee, and the parachute tangling.
What should change?
Separate investigation observations:
Cup depth 2 cm: Payload left during climb in 4 of 5 flights.
Cup depth 4 cm: Payload stayed attached in 3 of 5 flights.
Cup depth 3 cm: Payload released near apogee in 5 of 5 flights.

A. A small electronic timer

B. A rubber band ejector

C. A loose nose cup that releases as the rocket slows at apogee

D. A spring that pushes the parachute out

Correct: C

3. (2 points) Why does the pressure-height curve flatten at high pressure?

Unit: Apogee, ceiling height, and choosing pressure.
Worked-case data: Your chart shows 50 psi -> 17 ft, 60 psi -> 21 ft, 70 psi -> 24 ft, with about ±1.5 ft spread. The ceiling is 25 ft. Which pressure should you choose?
Separate investigation observations:
Chart: 45 psi -> 15 ft, 55 psi -> 19 ft, 65 psi -> 22 ft.
Spread: About ±1 ft at each pressure.
Venue: Ceiling announced at 21 ft.

A. The bottle holds less air

B. Timers stop earlier

C. Drag grows rapidly with speed

D. Gravity gets stronger with height

Correct: C

4. (2 points) Which phase of the flight earns most of the time aloft?

Unit: How an air-powered bottle rocket climbs.
Worked-case data: A rocket reaches apogee 1.2 s after launch, and the payload then descends for 11.5 s. What share of time aloft came from the descent?
Separate investigation observations:
40 psi: Peak about 12 ft; time aloft 8.1 s.
55 psi: Peak about 18 ft; time aloft 12.4 s.
70 psi: Peak about 24 ft; time aloft 16.0 s in a 30 ft hall.

A. The coasting climb

B. The time on the launcher

C. The parachute descent

D. The powered phase

Correct: C

5. (2 points) Time aloft plotted against peak height forms a straight line. What does the slope tell you?

A. Seconds of descent gained per foot of height

B. The launch pressure

C. The ceiling height

D. The rocket’s mass

Correct: A

6. (4 points) Explain why a parachute reaches a steady falling speed.

Unit: Parachute physics: drag and slow descent.
Worked-case data: A 4.7 g payload uses a 40 cm diameter dome (C_d about 1.5). Estimate its descent speed and the time to fall 7 m.
Separate investigation observations:
30 cm canopy: Opens every time; 7.8 s descent.
60 cm canopy: Opens 6 of 10 times; 14.9 s when open, 3 s when not.
50 cm canopy, looser fold: Opens 10 of 10; 12.6 s descent.

answer: As it speeds up, drag grows with the square of speed until it equals the weight; then the net force is zero and speed stays constant. (4 points).

7. (4 points) Describe how to check that a nose meets the rules.

Unit: Building a legal, stable rocket.
Worked-case data: A rocket's CG is 18 cm from the nose and its CP is 21 cm from the nose. The bottle diameter is 8 cm. Is it stable enough?
Separate investigation observations:
Flight 1: Rocket corkscrews and lands near a wall.
Inspection: One fin is twisted about 8° from the body line.
Flight 2: Fin realigned: rocket flies straight up.

answer: Place a standard bottle cap over the tip; the nose is legal if it does not touch the inside top of the cap. (4 points).

8. (4 points) What two consequences follow from not having a practice log?

Unit: The event and its score: time aloft and multipliers.
Worked-case data: A team with a complete log records 14.20 s on launch 1 (clean) and 19.80 s on launch 2, but launch 2 grazed a light on the way up. What is its score?
Separate investigation observations:
Ceiling: 24 ft gym.
Log data: 60 psi: about 20 ft peak, 15 s aloft; 75 psi: about 26 ft peak.
Team plan: Launch at 75 psi for "more height."

answer: The score is multiplied by 0.70, and the team is limited to a supervisor-chosen safe pressure. (4 points).

9. (4 points) What happens if the payload releases too early?

Unit: Deployment and separation.
Worked-case data: Video shows a payload leaving the rocket 0.3 s after launch, well before apogee, and the parachute tangling. What should change?
Separate investigation observations:
Cup depth 2 cm: Payload left during climb in 4 of 5 flights.
Cup depth 4 cm: Payload stayed attached in 3 of 5 flights.
Cup depth 3 cm: Payload released near apogee in 5 of 5 flights.

answer: The parachute opens at high speed, risking tears or tangles, and starts its descent below the rocket's apogee. (4 points).

10. (4 points) Give one source of error in angle measurements and one way to reduce it.

answer: Missing the exact peak moment or drift away from the launcher; reduce with video, two observers, or averaging several flights. (4 points).

11. (6 points) Solve this case and explain your reasoning: A rocket reaches apogee 1.2 s after launch, and the payload then descends for 11.5 s. What share of time aloft came from the descent?

reasoning: Total time aloft = 1.2 + 11.5 = 12.7 s.
Descent share = 11.5 ÷ 12.7 about 0.91.
About 91 percent of the score comes from the parachute descent. (3 points).

conclusion: Climb only needs to be high enough; descent rate is where most of the time is won. (3 points).

12. (6 points) Solve this case and explain your reasoning: A log has 12 flights recording pressure, height, time, and fin count. Is it complete?

reasoning: Required: pressure, height, time - all present.
Additional parameters: only fin count, one of the two needed.
Total parameters: four, below the five required. (3 points).

conclusion: The log is incomplete (x 0.85 and a capped pressure); adding parachute diameter for every flight would fix it. (3 points).

13. (2 points) Which parachute should the team choose?

Supplied synthetic observations:
30 cm canopy: Opens every time; 7.8 s descent.
60 cm canopy: Opens 6 of 10 times; 14.9 s when open, 3 s when not.
50 cm canopy, looser fold: Opens 10 of 10; 12.6 s descent.

A. The 50 cm canopy with the looser fold

B. The 60 cm canopy

C. The 30 cm canopy

Correct: A

14. (2 points) Define “Center of gravity (CG)” in the context of Ping-Pong Parachute.

answer: The rocket’s balance point. (2 points).

15. (2 points) Define “Parachute payload system” in the context of Ping-Pong Parachute.

answer: A ping pong ball with its taped-on parachute. (2 points).

16. (2 points) Define “Passive deployment” in the context of Ping-Pong Parachute.

answer: Release driven by the rocket’s own motion, without stored energy. (2 points).

17. (2 points) Define “Canopy area” in the context of Ping-Pong Parachute.

answer: The projected area of the open parachute. (2 points).

18. (2 points) Define “Cap test” in the context of Ping-Pong Parachute.

answer: Check that a blunt nose does not touch the inside top of a bottle cap. (2 points).